

# Authorization Non-Amplification under Chosen-Context Signer Harvesting

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## Abstract

A signature limits who could have produced a message. It does not limit how many consequential actions that message can unlock. This distinction is easy to miss in authorization systems whose collector may solicit genuine signatures on adaptively chosen requests, retain the resulting artifacts, and present them to multiple executors.

We study *authorization non-amplification* (ANA), a per-issued-instance trace property connecting cryptographic approval events to provider-entry cardinality. An entry must map to one exact issued context and, for every required key not yet revealed, to a prior signature on that context’s exact slot message. No issued instance may map to more than one entry. ANA is not implied by EUF-CMA security or by signer-to-collector Lowe-style injective agreement: replaying one valid artifact preserves both while violating ANA with probability one.

We define a finite multi-user chosen-context experiment and give a compiler from an EUF-CMA signature scheme, a collision-resistant hash, an injective typed encoding, and ideal issue-and-consume and mediation resources. An unwitnessed entry yields a signature forgery or hash collision; duplicate entry is excluded by the linear resources. A real-resource refinement bound exposes registry and mediation failures as separate trace events rather than cryptographic advantages. Four Tamarin models provide symbolic case studies and nine deliberately weakened comparisons. The result concerns admission, not human identity, display fidelity, policy wisdom, or exactly-once physical effects.

## 1 Introduction

Digital signatures answer a message-authenticity question: did a holder of a particular key sign these bytes? Consequential authorization adds a cardinality question: how many provider entries may those bytes authorize? The second answer is not contained in the first.

Consider a service that asks an enrolled approver to sign a transfer, release, deployment, or control action. The service may be honest, but the security boundary should survive a compromised collector. Such a collector can request many genuine signatures, choose the order in which requests are shown, retain old artifacts, combine signatures obtained at different times, and race the same artifact through several executors. None of those operations requires a signature forgery.

This paper studies that collector as the adversary. The adversary chooses the semantic authorization context, subject to a closed typed grammar, and obtains genuine signatures from uncompromised pinned keys. It also controls transport, storage, scheduling, and presentation. The intended guarantee is deliberately narrow: if an executor crosses a provider boundary, the exact admitted action must have one prior issued authorization instance and, for each then-unrevealed required key, a prior signing event on the exact slot message; one issued instance must not account for two entries.

We call the guarantee *authorization non-amplification* (ANA). For a trace  $T$ , let

$$\text{amp}_T(d, u) = |\{e : e = \text{Entry}(d, u, h, \tau, c, S) \text{ occurs in } T\}|.$$

The cardinality clause is  $\text{amp}_T(d, u) \leq 1$  for every issued instance key  $(d, u)$ ; this is replay prevention at the provider-entry boundary. The witness clause is separate: it relates each entry to one exact trusted issuance record and to every exact slot-message signing event required by the policy recorded in that issuance.

The contribution is a finite multi-user chosen-context experiment in which the adversary wins by causing an unwitnessed provider entry or by using one issued instance for two entries. We prove a compiler in an ideal-resource model and a refinement bound that exposes concrete registry and mediation failures. ANA is not a new correspondence logic. It is a security game for a particular cross-role correspondence—from issuance and signer events to executor entry—under chosen-context collection, adaptive reveal, and concurrent admission. The proof separates three obligations that are often collapsed:

1. **Message authenticity.** Every unrevealed required key signed the exact accepted typed context.
2. **Authorization-instance coherence.** Every required signature and the executor refer to the same independently issued context and pinned policy. This does not claim that approvers were contemporaneous or mutually aware.
3. **Non-amplification.** The trusted issue-and-consume record that witnesses one provider entry cannot witness another.

The distinction is cryptographically useful. EUF-CMA counts previously unsigned messages; ANA counts admitted actions. Lowe’s injective agreement is parameterized by roles and agreed data and relates protocol runs between peers [2]; ANA uses standard correspondence machinery but relates authorization evidence to an executor’s provider-entry events. Section 5 proves that EUF-CMA and a signer-to-collector agreement claim do not supply the missing entry bound.

The paper makes five concrete contributions:

- a finite multi-user, chosen-authorization-context experiment with explicit issuance, signing, reveal, verification, consumption, and provider-entry events;
- a witness-based ANA definition that combines exact semantic coverage, pinned-policy fidelity, and an injective map from entries to issued instances;
- a compiler from standard signatures and hashes plus an ideal exact issue-and-consume resource and a one-use mediator;
- a reduction whose multi-user sum ranges over every enrolled challenge key, not merely the keys selected by a winning policy, together with exact query and runtime accounting; and
- separation results and machine-checked case studies showing why a random nonce, an authentic signature, or signer-to-collector injective agreement does not by itself make an authorization single-use at the provider boundary.

The strongest claims stop at provider entry. A provider may fail, time out, or perform an effect whose acknowledgement is lost. ANA therefore forbids a blind retry after an indeterminate entry, but it does not claim exactly-once physical execution. It also does not prove that a key belongs to a particular person or that a display faithfully represented the signed action.

## 2 Setting and Syntax

### 2.1 Signature and hash assumptions

Let  $\text{Sig} = (\text{Kg}, \text{Sign}, \text{Verify})$  be a signature scheme. For each enrolled index  $j \in [N]$ , the challenger samples  $(sk_j, pk_j) \leftarrow \text{Kg}(1^\lambda)$ . We use standard existential unforgeability under adaptive chosen-message attack and write  $\text{Adv}_{\text{Sig}}^{\text{euf-cma}}(\mathcal{B})$  for the success probability of a forger  $\mathcal{B}$  [1].

Let  $H : \{0, 1\}^* \rightarrow \{0, 1\}^\kappa$  be collision resistant. We write  $\text{Adv}_H^{\text{cr}}(\mathcal{C})$  for the probability that  $\mathcal{C}$  outputs two different inputs with the same digest.

## 2.2 Typed authorization language

An action  $a$  is a value in a closed typed action grammar. Its encoding  $\text{Enc}_A(a)$  is deterministic and injective. A parser rejects unknown members, missing required members, duplicate members, invalid types, non-canonical numbers, and invalid text encodings. A transport such as deterministic CBOR or JCS can implement the wire representation, but the security premise is the injectivity of the complete typed profile, not the name of the transport.

A pinned-slot policy is an ordered vector of length  $1 \leq t \leq N$ ,

$$\Pi = ((s_1, z_1, pk_{j_1}), \dots, (s_t, z_t, pk_{j_t})),$$

where slot labels  $s_k$ , enrolled principal identifiers  $z_k$ , and keys  $pk_{j_k}$  are pairwise distinct and every pair  $(z_k, pk_{j_k})$  belongs to the certified enrollment relation defined below. This paper proves the all-of- $t$  policy. It does not silently generalize the theorem to arbitrary threshold or monotone access structures.

Let  $p$  be an immutable policy-version commitment,  $i$  the initiating principal,  $r$  the relying-party audience,  $d$  the enforcement domain, and  $q$  a  $\lambda$ -bit instance chosen without replacement by a trusted issuer for the lifetime of the experiment. Define

$$h = H(\text{Enc}(\text{ANA-ACTION-v1}, \text{Enc}_A(a))), \quad c = (d, h, p, i, r, \Pi, q).$$

The context encoding  $\text{Enc}_C(c)$  and the outer message encoding below are deterministic and injective on their closed grammars. No slot nonce is needed:  $q$  gives instance uniqueness and  $s_k$  separates slots. The signed message for slot  $s_k$  is

$$m(c, s_k) = \text{Enc}(\text{ANA-SIGN-v1}, c, s_k).$$

The admission key is

$$u = H(\text{Enc}(\text{ANA-ADMIT-v1}, c)).$$

Including  $d$  in  $c$  prevents a signature gathered for one enforcement domain from crossing into another. Including  $\Pi$  makes the complete set of pinned slots and keys part of every signature. Including  $i$  permits an executor to enforce separation against the issuer-recorded initiator identifier rather than against a mutable label.

## 2.3 Certified-key model

The challenger fixes the certified enrollment relation  $\text{Cert} = \{(z_j, pk_j) : j \in [N]\}$  before the game. Principal identifiers and canonical public-key encodings are unique in this relation; key generation is repeated if it produces a duplicate encoding. The challenger rejects a policy containing a pair outside  $\text{Cert}$ , a duplicate key, duplicate identifier, or duplicate slot. This prevents a collector from relabeling  $pk_j$  as another enrolled principal. Enrollment, proof of possession, and the binding from an identifier to a natural person are outside the theorem. The result speaks about distinct enrolled keys and identifiers, not distinct human wills.

The challenger also fixes a trusted audience function  $\text{Aud}$  before play.  $\text{Aud}(d)$  is the authenticated relying-party identity expected at enforcement domain  $d$ . The collector cannot modify this value. An issued context may carry any audience selected through the chosen-context interface, but it is admissible at  $d$  only when its signed  $r$  equals  $\text{Aud}(d)$ .

The experiment lets the adversary choose the field  $i$  in an issuance query and therefore proves only that no approver identifier equals the *issuer-recorded*  $i$ . A deployment claiming initiator–approver separation must authenticate the initiating principal and require the trusted issuer to set  $i$  from that authenticated identity. It must not copy  $i$  from an untrusted collector.

An adversary may reveal  $sk_j$ . Authenticity for key  $j$  ends at the first reveal event. A later reveal does not alter a previously ordered trace prefix. This is a weak-corruption model, not forward security or offline anti-backdating.

### 3 Exact Issue, Consume, and Mediation Resources

The cryptographic proof uses two explicit stateful resources. Writing them down prevents a verbal “single-use” claim from entering the theorem as an unstated database assumption.

#### 3.1 Exact issue-and-consume resource

For each domain  $d$ , resource  $\mathcal{F}_{IC}$  stores a durable map  $R_d$ . Each value contains the complete immutable context, not merely a status bit.

```

Register(d, u, c):
  if u has ever appeared in R_d: return false
  R_d[u] <- (context=c, status=unused)
  emit Registered(d,u,c)
  return true

Lookup(d, u):
  return R_d[u] or bottom

ConsumeExact(d, u, c):
  if R_d[u] != (context=c, status=unused): return bottom
  allocate a never-before-used ideal handle tau
  R_d[u].status <- consumed
  emit Consumed(d,u,c,tau)
  return tau

```

Calls are linearizable and durable. No transition recreates **unused**. A successful consume is therefore both an exact issuance lookup and a one-time state transition. The resource refuses a caller that presents the right  $u$  with different context bytes.

#### 3.2 One-use provider mediator

Resource  $\mathcal{F}_{MED}$  is the only interface that can emit a provider-entry event. On input  $(d, u, c, S, \tau)$  it checks for one unmatched prior  $\text{Consumed}(d, u, c, \tau)$  event and parses action digest  $h$  from  $c$ . It marks  $\tau$  used, emits  $\text{Entry}(d, u, h, \tau, c, S)$  once, and refuses every other call. Carrying the accepted evidence set  $S$  in the trace event makes the security predicate well defined; it need not be sent to the external provider. This is complete mediation at the modeled boundary.

The resource models entry, not effect. A successful call means that the request crossed the boundary. It says nothing about what the provider later did.

## 4 Security Experiment

### 4.1 Oracle interface and budgets

Experiment  $\text{Exp}_{\mathcal{A}}^{\text{ANA}}(1^\lambda)$  fixes  $N$  enrolled keys and gives their public descriptions to  $\mathcal{A}$ . The adversary makes at most  $Q_I$  issuance queries,  $Q_{S,j}$  signing queries to key  $j$ ,  $Q_A$  admission queries, and  $Q_R$  reveal queries. It controls their interleaving and may invoke admission concurrently.

The challenger provides four oracles.

**Issue** $(d, a, p, i, r, \Pi)$ .

Validate the closed action, require every policy pair to be in **Cert**, and refuse if an approver identifier equals  $i$ . Sample  $q$  uniformly from the issuer’s as-yet-unused values in  $\{0, 1\}^\lambda$ , construct  $c$  and  $u$ , and call  $\mathcal{F}_{IC}.\text{Register}(d, u, c)$ . Refuse if the instance space is exhausted or registration fails. Emit **Issued** $(d, u, c, a)$  and return  $(u, c)$ .

**SignCtx** $(j, d, u, s)$ .

Look up  $(d, u)$  and require an unused exact context  $c$  whose policy assigns  $(s, z_j, pk_j)$  to slot

$s$ . Compute  $m(c, s)$ , emit  $\text{Signed}(j, d, u, c, s, m(c, s))$ , and return  $\text{Sign}(sk_j, m(c, s))$ . Thus the adversary chooses every semantic field except the issuer's fresh  $q$ , may obtain signatures on arbitrarily many chosen issued contexts, and may later discard or splice the artifacts.

**Reveal( $j$ ).** Emit  $\text{Revealed}(j)$  and return  $sk_j$ . Repeated calls have no additional effect.

**Admit( $d, u, a, S$ ).**

Run the construction in Section 6, where  $S$  is the set of presented slot signatures and the executor obtains its expected audience from the trusted function  $\text{Aud}(d)$ . Return its result.

Malformed queries are refused without emitting a security event. The challenger records a totally ordered trace of oracle calls and resource linearization points. This ordering is part of the experiment; the paper does not claim an offline verifier can reconstruct it after key compromise.

## 4.2 Entry witness

Let  $e = \text{Entry}(d, u, h, \tau, c, S)$  occur at trace position  $t_e$ . The event has an *exact authorization witness* if there is a unique prior  $\text{Issued}(d, u, c, a)$  event satisfying all of the following:

1.  $c = (d, H(\text{Enc}(\text{ANA-ACTION-v1}, \text{Enc}_A(a))), p, i, r, \Pi, q)$  and its action digest equals  $h$ ;
2. for every  $(s_k, z_k, pk_{j_k}) \in \Pi$  whose key was not revealed before  $t_e$ , a prior event  $\text{Signed}(j_k, d, u, c, s_k, m(c, s_k))$  exists;
3.  $S$  contains exactly one pair  $(s_k, \sigma_k)$  for every slot in  $\Pi$ , each  $\sigma_k$  verifies under its pinned key on the exact message  $m(c, s_k)$ , the signed audience satisfies  $r = \text{Aud}(d)$ , the policy entries are pairwise distinct, and no  $z_k$  equals the signed initiator  $i$ ; and
4. one prior  $\text{Consumed}(d, u, c, \tau)$  event exists and  $\tau$  witnesses no other provider entry.

The witness does not say that the approvers met, signed simultaneously, knew about one another, or perceived a faithful display. It proves common authorization-instance coherence over  $c$ .

**Definition 1** (Authorization non-amplification). *Adversary  $\mathcal{A}$  wins the ANA experiment if either:*

1. *an entry occurs without an exact authorization witness; or*
2. *two distinct entries in the same enforcement domain contain the same admission key  $u$ .*

*Its advantage is  $\text{Adv}_{\mathcal{A}}^{\text{ana}} = \Pr[\text{Exp}_{\mathcal{A}}^{\text{ANA}} = 1]$ .*

This definition intentionally gives no authenticity guarantee for a slot whose secret key was revealed before entry. The second, cardinality clause remains in force regardless of key reveal.

## 5 Separations from Standard Notions

These results locate the new property. They do not claim that EUF-CMA or injective agreement is weak; they show that those notions answer different questions.

**Proposition 1** (Stateless replay impossibility). *Let a complete authorization system expose a stateless admission algorithm whose acceptance of a valid artifact does not change any state shared with a later call. If the same artifact can reach the provider boundary twice, then the system cannot satisfy ANA. For a deterministic complete verifier, the ANA adversary wins with probability one and produces no signature forgery.*

*Proof.* Obtain one valid artifact through the system's issuance and signing interfaces, then submit it to two admission calls. Statelessness leaves the second call in the same relevant verification state as the first, so completeness makes both calls accept and emit entries. Every signature was returned by an authorized signing interface, yet two entries use the same issued instance. Thus EUF-CMA, even with perfect message authentication, cannot supply the missing entry bound. A probabilistic verifier with independent fresh coins and one-call acceptance probability  $p$  yields the same violation with probability at least  $p^2$ .  $\square$

**Proposition 2** (A signed random nonce is insufficient). *Adding a fresh nonce to the signed message does not imply ANA if executors do not share an atomic consume transition for that nonce.*

*Proof.* Replay the complete message, nonce, and genuine signature to two stateless executors. Freshness distinguishes the message from other messages; it does not make either executor learn that the other has accepted it.  $\square$

**Proposition 3** (Byte unforgeability does not imply semantic coverage). *If two distinct typed contexts  $x \neq x'$  have the same accepted signed-byte encoding, an adversary wins the witness clause using one genuine signing query and no EUF-CMA forgery.*

*Proof.* Ask the signer to sign the common byte string while the semantic record is  $x$ , then present the signature with  $x'$ . Verification succeeds on the queried bytes, but there is no exact prior semantic signing event for  $x'$ .  $\square$

**Proposition 4** (Signer-to-collector injective agreement does not imply ANA). *Lowe-style injective agreement between an approver and a collector does not imply ANA at an executor boundary.*

*Proof.* Take any protocol in which each completed collector run has a unique matching approver run on data  $c$ . After the collector completes, wrap its accepted transcript in an executor API that emits a provider entry on every call. Replaying that transcript to the API does not create a second collector completion or a second approver run, so the original injective- agreement relation remains true. The wrapper nevertheless emits two entries for one authorization and violates ANA.  $\square$

**Proposition 5** (ANA does not imply agreement among approvers). *ANA does not imply mutual or injective agreement among the approvers.*

*Proof.* Approvers may sign sequentially through the chosen-context oracle without communicating or knowing one another. The issue-and-consume resource can still give the resulting entry an exact injective witness. ANA therefore establishes neither mutual awareness nor a peer-to-peer agreement run.  $\square$

The executor event is the missing endpoint in the first construction. ANA can be written as an injective correspondence from provider entries to issue, consume, and signing events. The chosen-context experiment turns that cross-role correspondence into a computational target and identifies the state and mediation resources needed to realize it.

## 6 Construction

The following compiler realizes the experiment for any all-of- $t$  pinned-slot policy.

```

Issue(d, a, p, i, r, Pi):
  validate the closed action grammar and certified pinned-slot policy
  require 1 <= |Pi| <= N
  reject if an approver id in Pi equals initiator id i
  q <- uniform unused value in the issuer's {0,1}^lambda instance set
  h <- H(Enc("ANA-ACTION-v1", Enc_A(a)))
  c <- (d, h, p, i, r, Pi, q)
  u <- H(Enc("ANA-ADMIT-v1", c))
  reject unless F_IC.Register(d, u, c) = true
  emit Issued(d, u, c, a)
  return (u, c)

Approve(sk_j, d, u, c, s_j):
  reject unless F_IC.Lookup(d, u) = (c, unused)
  reject unless Pi in c assigns (s_j, z_j, pk_j)
  m_j <- Enc("ANA-SIGN-v1", c, s_j)
  sigma_j <- Sign(sk_j, m_j)

```

```

    emit Signed(j,d,u,c,s_j,m_j)
    return (s_j,sigma_j)

Admit(d, u, a, S):
  (c,status) <- F_IC.Lookup(d,u); reject unless status = unused
  parse c as (d_c,h,p,i,r,Pi,q) under the closed grammar
  reject unless d_c = d
  reject unless r = Aud(d)
  reject unless h = H(Enc("ANA-ACTION-v1",Enc_A(a)))
  reject unless Pi has unique slots, ids, and certified pinned keys
  reject if an approver id in Pi equals initiator i
  for each (s_j,z_j,pk_j) in Pi:
    require exactly one (s_j,sigma_j) in S
    m_j <- Enc("ANA-SIGN-v1",c,s_j)
    reject unless Verify(pk_j,m_j,sigma_j) = true
  tau <- F_IC.ConsumeExact(d,u,c); reject if tau = bottom
  return F_MED.Enter(d,u,c,S,tau)

```

The executor constructs  $a$  from the operation it is about to submit. It does not accept a descriptive action label as a substitute for the provider request. The policy commitment  $p$  identifies an immutable policy version. The audience  $r$  prevents a signature collected for one relying party from being presented as authority for another. The complete policy  $\Pi$  is signed, so a collector cannot add, remove, or replace a required slot after gathering signatures. When a deployment uses  $i$  for separation of duties, the trusted issuer must derive it from the authenticated initiating principal as described above.

The construction places validity windows and clocks outside the core theorem. A deployment may add a typed validity object to  $c$  and refuse it under a defined trusted-time policy. Merely writing a timestamp into  $c$  would not prove clock correctness or anti-backdating, so the core claim does not depend on one.

## 7 Security Analysis

**Lemma 1** (Exact-context binding). *Conditioned on no collision in  $H$ , equality of two valid slot-message byte strings implies equality of their enforcement domain, action, policy commitment, initiator, audience, pinned-slot policy, authorization instance, and slot.*

*Proof.* Injectivity of the outer closed encoding gives equality of the literal domain label, both contexts, and the slot. Equality of the contexts gives equality of every directly encoded field and of the action digests. If the canonical action encodings differed, the equal digests would be a collision in  $H$ .  $\square$

**Lemma 2** (Functional completeness). *Assume signature correctness and the ideal resources. If `Issue` returns  $(u,c)$ , every slot in the nonempty policy is approved through `SignCtx` while the record remains unused, and `Admit` is called with the issued action and those signatures at domain  $d$  with the issued  $r = \text{Aud}(d)$ , then the first admission emits one provider entry. Every later admission for the same  $(d,u)$  is refused.*

*Proof.* Signature correctness makes every pinned verification succeed. The issued action reproduces the digest in  $c$ , and the certified policy satisfies the deterministic structural checks. The first exact consume therefore returns a fresh mediator token, which the mediator accepts once. The consume changes the record to `consumed`, so a later lookup or consume cannot reach entry.  $\square$

**Lemma 3** (Ideal non-amplification). *With  $\mathcal{F}_{\text{IC}}$  and  $\mathcal{F}_{\text{MED}}$ , two provider entries in one domain cannot contain the same admission key  $u$ .*

*Proof.* Every entry consumes one fresh mediator token produced by a successful `ConsumeExact` $(d,u,c)$ . Linearizability totally orders consumes for  $u$ . The first success changes the unique exact record from `unused` to `consumed`; no rule recreates `unused`. A second consume for  $u$  cannot succeed. The mediator also refuses reuse of the first token.  $\square$

**Theorem 1** (ANA compiler). *Let  $\text{Sig}$  be EUF-CMA secure,  $H$  collision resistant, and the declared typed encodings injective. For every probabilistic polynomial-time adversary  $\mathcal{A}$  using at most  $N$  enrolled keys,  $Q_I$  issuance queries,  $Q_{S,j}$  signing queries to key  $j$ ,  $Q_A$  admission queries, and  $Q_R$  reveal queries, there are collision finder  $\mathcal{C}_H$  and EUF-CMA forgers  $\mathcal{B}_1, \dots, \mathcal{B}_N$  such that*

$$\text{Adv}_{\mathcal{A}}^{\text{ana}} \leq \text{Adv}_H^{\text{cr}}(\mathcal{C}_H) + \sum_{j=1}^N \text{Adv}_{\text{Sig}}^{\text{euf-cma}}(\mathcal{B}_j).$$

*Forger  $\mathcal{B}_j$  makes at most  $Q_{S,j}$  signing-oracle queries. Its simulation generates  $N-1$  key pairs, locally answers at most  $\sum_{\ell \neq j} Q_{S,\ell}$  signing queries, performs at most  $NQ_A$  signature verifications, and adds polynomial encoding, hashing, policy, and table work in the stated query bounds. The sum is over the complete enrolled key universe fixed before play, not a policy selected after a win.*

*Proof.* We use four games.

*Game  $G_0$ .* Run the real ANA experiment with the ideal state resources.

*Game  $G_1$ .* Domain-separate the action and admission-key inputs as in the construction and record every input on which the challenger evaluates  $H$ . Abort if two distinct recorded inputs have the same digest. A challenger that outputs the first such pair is  $\mathcal{C}_H$ , so

$$|\Pr[G_0 = 1] - \Pr[G_1 = 1]| \leq \text{Adv}_H^{\text{cr}}(\mathcal{C}_H).$$

*Game  $G_2$ .* For each  $j \in [N]$ , abort if an entry's evidence set contains a required valid signature under  $pk_j$ , no reveal of  $j$  precedes the entry, and the signing table contains no exact prior row  $(j, d, u, c, s, m(c, s))$  for that accepted slot. Construct  $\mathcal{B}_j$  by embedding its EUF-CMA challenge key as  $pk_j$ , generating every other key, and forwarding each  $\text{SignCtx}$  query for  $j$  to its signing oracle. It simulates issuance, other signers, the resources, and all public checks exactly. A reveal of  $j$  ends the experiment's authenticity claim for that key, so the fixed- $j$  reduction may abort on that reveal without losing a winning event to which the definition applies.

If the  $G_2$  bad event occurs first, the accepted signature verifies on message  $m^* = m(c^*, s^*)$ . No exact signing row exists. A query on the same bytes under a different semantic context would contradict exact-context binding in  $G_1$ . Thus  $m^*$  was not submitted to the EUF-CMA oracle, and the accepted pair is a forgery. A union bound over the key universe gives

$$|\Pr[G_1 = 1] - \Pr[G_2 = 1]| \leq \sum_{j=1}^N \text{Adv}_{\text{Sig}}^{\text{euf-cma}}(\mathcal{B}_j).$$

*Game  $G_3$ .* Associate every accepted unrevealed slot with its exact prior signing-table row. This changes no probability. The exact issue lookup fixes one context  $c$  for  $u$ ; outer-encoding injectivity and the  $G_1$  condition make the accepted action and every signed field equal to that context. The deterministic checks enforce the trusted audience, unique pinned keys and slots, and exclusion of the signed initiator. Hence every entry has the required cryptographic witness.

Finally, ideal non-amplification rules out two entries with the same  $(d, u)$ . No ANA winning condition remains in  $G_3$ , so  $\Pr[G_3 = 1] = 0$ . Summing the game differences proves the bound. The stated query and runtime costs follow directly from the simulation.  $\square$

## 7.1 Concrete interpretation

If every per-key EUF-CMA advantage is at most  $\epsilon_{\text{sig}}$  and the hash-collision advantage is at most  $\epsilon_H$ , the theorem gives

$$\text{Adv}_{\mathcal{A}}^{\text{ana}} \leq N\epsilon_{\text{sig}} + \epsilon_H.$$



The  $N$  factor is deliberately conservative. It avoids a post-selected two-key bound when an adversary can range over many enrolled keys and policies.

The theorem is not a durability proof for a database or a code-level proof of complete mediation. The next corollary states exactly what a real deployment must add.

## 7.2 Real-resource refinement

Let  $R$  and  $M$  be probabilistic polynomial-time implementations of the issue-and-consume and mediation interfaces. Experiment  $\text{Exp}_{\mathcal{A},R,M}^{\text{ANA},\text{real}}(1^\lambda)$  is the ANA experiment of Section 4 with  $R$  and  $M$  substituted for the ideal resources. It records their registration, consume, and provider-entry events in the same ordered trace and applies the same ANA winning predicate. Write  $\text{Adv}_{\mathcal{A},R,M}^{\text{ana},\text{real}}$  for its winning probability. In that experiment, define:

- **RegistryBreak** to occur if a successful consume lacks one exact prior registration  $(d, u, c)$ , if one  $u$  is registered for two different contexts, or if two consumes for the same  $u$  both succeed; and
- **MediationBreak** to occur if a provider entry lacks one unmatched prior successful consume token bound to the same  $(d, u, c)$  with the same action digest  $h$ , or if one token witnesses two entries.

These are trace events in a stated experiment, not undefined labels for “something went wrong.” Their probabilities must be justified by a refinement proof, transactional implementation argument, or measured fault model for the deployment.

**Corollary 1** (Deployment composition). *Replacing the ideal resources by implementations  $R$  and  $M$  gives*

$$\text{Adv}_{\mathcal{A},R,M}^{\text{ana},\text{real}} \leq \text{Adv}_H^{\text{cr}}(C_H) + \sum_{j=1}^N \text{Adv}_{\text{Sig}}^{\text{euf-cma}}(\mathcal{B}_j) + \frac{\Pr_{\text{Exp}_{\mathcal{A},R,M}^{\text{ANA},\text{real}}}[\text{RegistryBreak}]}{\Pr_{\text{Exp}_{\mathcal{A},R,M}^{\text{ANA},\text{real}}}} + \frac{\Pr_{\text{Exp}_{\mathcal{A},R,M}^{\text{ANA},\text{real}}}[\text{MediationBreak}]}{\Pr_{\text{Exp}_{\mathcal{A},R,M}^{\text{ANA},\text{real}}}}.$$

*Proof.* Conditioned on neither implementation event, the observed issue, consume, and entry trace satisfies the two ideal-resource safety relations used in the compiler theorem. Apply that theorem and then a union bound over the two implementation events.  $\square$

The corollary keeps operational correctness visible. It would be misleading to call either implementation probability negligible merely because the signatures are cryptographically strong.

## 8 Trace-Prefix Authenticity and Backdating

**Lemma 4** (Trace-prefix authenticity). *Suppose an entry whose evidence set contains a required valid signature under  $pk_j$  occurs at position  $t_e$ , no reveal of  $sk_j$  precedes  $t_e$ , and a reveal occurs later. Except with probability bounded by the corresponding EUF-CMA and collision advantages, the exact signing event for the accepted slot precedes  $t_e$ .*

*Proof.* Stop the experiment at  $t_e$ . The later reveal is outside the prefix and does not change its distribution. The  $G_1$ – $G_2$  argument from the compiler theorem then gives the exact prior signing event.  $\square$

**Proposition 6** (Ordinary signatures do not prove offline anti-backdating). *After learning  $sk_j$ , an adversary can create a valid signature on a message containing an earlier claimed timestamp. An offline verifier given only  $(m, \sigma, pk_j)$  cannot distinguish it from a signature produced before compromise.*

*Proof.* The adversary runs the honest signing algorithm after compromise. The timestamp is only signed data; it is not an independently ordered event. Both signatures satisfy the same verification relation.  $\square$

Forward-secure signatures or an independently witnessed append-only checkpoint can support a stronger statement. Neither is part of the ANA compiler.

## 9 Symbolic Case Studies

Four public Tamarin theories [34] test concrete single-approval, 2-of-2, and composed instances against a Dolev–Yao adversary that controls the network and can solicit honest signatures on chosen authorization contexts. The theories are not a substitute for the computational theorem: they idealize signatures and typed encoding, and their linear facts assume the state transitions they represent.

The two focused models report the following exact machine results with Tamarin 1.10.0 and Maude 3.4:

```
executable_honest_receipt: verified (9 steps)
core_authenticity_uv_gated: verified (11 steps)
acceptance_prefix_integrity_after_later_reveal: verified (12 steps)
no_replay_across_actions: verified (11 steps)
injective_acceptance_with_consumption: verified (11 steps)
unchecked_acceptance_is_injective: falsified (11 steps)

executable_quorum: verified (13 steps)
quorum_requires_two_distinct_uv_gated_signatures: verified (27 steps)
initiator_cannot_self_approve: verified (4 steps)
no_single_signer_fills_quorum: verified (4 steps)
commit_requires_signature_over_that_action: verified (8 steps)
```

The deliberately false receipt lemma removes the linear consumption input and finds the expected trace: one genuine artifact is accepted twice. An earlier quorum variant omitted the initiator from the signed context. Its counterexample used an original initiator who was also an approver, then substituted a distinct unsigned initiator so the self-approval check passed. Binding the initiator into the signed context removes that trace. This corrects a common but important misdescription of the attack: the substituted initiator is not the approver.

The composed theories add exact action identity, current registry view, revocation state, issuer pinning, authority scope, profile pins, one-time consumption, and entry. Nine weakened comparisons produce these attack classes:

Table 1: Machine-found separations with ideal signatures unchanged.

| Comparison                | Removed check                | Counterexample consequence                     |
|---------------------------|------------------------------|------------------------------------------------|
| Focused double acceptance | linear consumption           | one genuine approval accepted twice            |
| Composed double entry     | shared consume fact          | one authorization reaches entry twice          |
| Stale-head laundering     | exact current registry head  | narrowed authority accepted under an old view  |
| Class downgrade           | minimum assurance class      | weaker evidence satisfies a stronger class     |
| Denial becomes authority  | denial non-authorization     | signed denial treated as permission            |
| Scope expansion           | authority-scope pin          | out-of-scope action admitted                   |
| Profile swap              | complete profile pin         | different verification profile substituted     |
| Phantom challenge         | registration before exposure | unregistered challenge accepted                |
| Action-key swap           | canonical action-key binding | evidence for one action keys entry for another |

The focused models correspond to the compiler’s exact-signature and non-amplification obligations. The composed models test additional EMILIA reliance checks, which are outside the computational theorem.

## 10 Relation to Prior Work

**Authentication and injective agreement.** Bellare and Rogaway’s matching-conversation model and Lowe’s hierarchy make session correspondence explicit; Lowe distinguishes aliveness, weak agreement, non-injective agreement, and injective agreement between protocol participants [3, 2]. ANA borrows the discipline of explicit correspondence events but changes the relation being counted: provider entries must map

injectively to trusted issued authorization instances. Proposition 4 shows a protocol can preserve signer-to-collector injective agreement while a downstream executor amplifies one completed run into several entries. Proposition 5 shows that ANA permits asynchronous signers who never establish mutual agreement. We do not claim a new general logic of correspondence assertions. The new object is the chosen-context authorization experiment and its exact executor-entry witness.

**Computational authorization proofs.** Wang et al. give a computationally sound mechanized analysis of TPM 2.0 authorization protocols [4]. Barbosa et al. give a modular computational analysis of WebAuthn and CTAP2 [5]; Guan et al. and Schrempf study FIDO2 symbolically, including a human role [6, 7]. Those works analyze the security of their authorization or authentication ceremonies. ANA assumes an enrolled signature interface and places the collector under chosen-context adversarial control. Its proof target also includes executor entry, and its failure measure is the number of entries attributable to one issued authorization.

**Agent authorization gateways and records.** OAP intercepts tool calls before execution and produces signed audit records [15]. aiAuthZ moves identity- and policy-bound authorization off the agent host and uses an atomic single-use nonce store [16]. AgentBound describes receipts binding an action to delegation, policy, and semantic artifacts [17]. Llambí-Morillas and Fernández-Fernández formalize principal, request, context, and policy binding and identify request-to-runtime execution binding as an open research problem [18]. These works establish that pre-action gateways, operation binding, and signed records predate ANA. ANA’s narrower cryptographic delta is a multi-user EUF-CMA/collision-resistance reduction and an exact multi-signer entry witness against a collector allowed to harvest chosen-context signatures.

**Transaction authorization and trusted display.** Van Herreweghen and Wille analyze which Internet-payment security requirements an EMV deployment does and does not meet [8]. “See what you sign” makes trusted input and output part of binding a signature to human intent [9]; Notary builds a trusted device for transaction approval [10]. ANA does not solve the display problem. It ensures that the executor cannot accept a different typed action from the one covered by the signature and cannot admit the same issued instance twice.

**Stateful authorization.** Formal analyses of PKCS#11 show that mutable state can be integral to a security claim [11]. StatefulAuth protects authorization policies that depend on request history [12]. Macaroons provide decentralized attenuation through contextual caveats [13]. ANA focuses on a different state relation: exact issue-and-consume state must injectively connect one signed authorization to one provider entry under concurrent presentation.

The need for spent-artifact state is classical rather than new. Online digital cash prevents double spending through a bank’s spent-coin state, while offline schemes detect it after deposit [19]. Linearizability supplies the concurrent-object semantics used by the ideal consume operation [21], and complete mediation supplies the reference-monitor principle behind the executor boundary [22]. ANA’s contribution is not the observation that replay needs state. It is the cryptographic experiment connecting malicious chosen-context collection, multi-signer typed evidence, exact issuance, and the provider-entry event.

Anonymous counting tokens let an issuer enforce a per-client issuance rate, including one token per client per message, while retaining the privacy goals of anonymous tokens [20]. That is an issuance-side cryptographic limit. ANA does not provide anonymity and assumes a trusted issuer has already created an instance; its question is whether a malicious collector can turn the exact multi-signer evidence for that instance into more than one provider entry.

SUDP gives an operation-bound, authenticator-backed single-use grant redeemed by a custodian so the agent never receives a reusable secret [23]. Krenn and Lorünser use smart-contract state to enforce a user-defined bound on delegated signature invocations [24]. Both demonstrate that single use can be built with a stateful redemption boundary. ANA differs in the adversary and witness: a malicious collector may harvest signatures for many issued contexts, and each executor entry must point to the exact multi-signer context and policy that survived that collection.

Lie and Marsh define ledger-authenticator unforgeability for reactive authorization judged against a finalized transcript, with adaptive corruption and a multi-user QROM bound [28]. ANA has no ledger, finality, or quantum-random-oracle claim. Its trace order comes from the experiment, and its later-reveal result is only prefix authenticity. In return, ANA makes multi-signer exact-action artifacts and the non-ledger executor admission event the object of the reduction.

Xu et al. identify *semantic replay*: a system can consume every token once and still issue fresh identifiers against the same underlying confirmation until its action-level budget is exceeded [14]. Their CapLease construction uses durable token-independent state to bound reissuance, admission, and recovery. That result is complementary and sets an important boundary on this paper. ANA is per issued instance  $(d, u)$ ; it does not stop a trusted issuer from creating a new  $(d, u')$  for semantically equivalent authority. ANA instead gives a computational chosen-context account of the evidence needed for one issued instance: exact issuer registration, prior signing events on every exact pinned-slot message for each then-unrevealed required key, and an injective map to provider entry. A system that needs a budget spanning several issued instances must compose ANA with a durable action- or confirmation-level ledger.

**Policy-aware and threshold cryptography.** Policy-Compliant Signatures make policy satisfaction part of an enhanced signature primitive, and attribute-based signatures hide signers behind attribute predicates [29, 30]. Bellare and Neven formalize multisignatures in which several certified public keys jointly authenticate one common document [25]; threshold signatures likewise can make several shares verify as one signature. ANA instead uses ordinary signatures under explicitly pinned keys and exposes each required slot. Replacing those signatures with a policy-aware or threshold primitive would not remove the need for exact issuance and one-time consumption at the executor boundary.

**Non-amplification terminology.** The phrase *non-amplification* has prior uses with different counted objects. Maffei, Mitchell, and Taly require “No Authority Amplification” as part of authority safety for object-capability isolation [26]. Xu et al. use *source-authority non-amplification* to prevent lossy agent memory consolidation from increasing the trust assigned to source information [27]. Neither property counts provider entries per issued authorization instance. We use ANA as a descriptive name for the trace property in Definition 1, not as a claim of priority over those terms.

**Concrete formats.** Compare lifts bit-string security arguments to typed protocol messages under verified format conditions [31]. ANA relies on the same lesson. Its transplantation separation explains why a byte-level signing oracle is insufficient when the application accepts two semantic contexts as the same bytes. This paper states typed-encoding injectivity as a premise; it does not claim a new verified parser framework.

**Logs and temporal evidence.** Transparency overlays and forward-secure audit logs can provide independently checkable ordering and post-compromise evidence [32, 33]. ANA’s trace-prefix lemma uses the experiment’s event order and makes no offline anti-backdating claim. A deployment that needs that stronger property must add one of those mechanisms.

## 11 Limitations and Ethics

The certified-key model does not prove that an enrolled key belongs to a named person or that two keys represent independent people. It proves distinct pinned credentials. Directory governance, identity proofing, collusion, coercion, and one-person-many-credentials attacks remain separate.

The collector may render action  $X$  while causing a signing device to sign the digest of action  $Y$ . ANA requires the admitted action to reproduce that signed digest and prevents the artifact from authorizing a different action  $Z$  absent a hash collision. The signature alone does not reveal  $Y$ , and the property cannot prove what the approver saw. High-consequence deployments need a trusted or independently authored display path.

The theorem covers a finite enrolled key universe and all-of- $t$  pinned-slot policies. It does not cover arbitrary threshold policies, dynamic key-directory updates, clock synchronization, policy authorship, or transparency-log consistency. It assumes the executor derives the canonical action from the operation it is about to submit.

The pair  $(d, u)$  is the unit counted by ANA. If an issuer creates a fresh  $(d, u')$  from the same human confirmation or an equivalent action, both instances may each be admitted once without violating ANA. Preventing that semantic replay requires durable state and a budget above the per-instance registry [14]; it is not implied by this compiler.

The work analyzes synthetic protocol traces and public formal artifacts. It does not probe third-party systems, collect personal data, or release an exploit against a deployed service. The negative models remove one check at a time to show why that check is load-bearing. Their publication lowers risk by making the assumptions auditable; it does not provide a turnkey compromise of an external system.

Most importantly, provider entry is not physical truth. An entry may have no effect, one effect, or an effect whose acknowledgement is lost. When the outcome is indeterminate, ANA keeps the authorization consumed and requires separate authenticated reconciliation. It does not authorize a blind retry.

## 12 Conclusion

Cryptographic authorization needs a unit of account. EUF-CMA counts new signed messages. Consequential systems also need to count admitted actions. ANA makes that second count explicit: each provider entry needs an exact issuance witness and, for each then-unrevealed required key, an exact-message signing-event witness; no issued authorization may witness two entries.

The proof assigns separate obligations to the cryptographic, encoding, state, and mediation layers. Signatures and collision resistance bind the exact semantic statement. A closed typed encoding prevents semantic transplant. Exact issue-and-consume state and complete mediation bound the number of entries.

matching signer run and still replay its output through a downstream executor. The core distinction is the role boundary. A collector may have one unique matching signer run and still replay its output through a downstream executor. matching signer run and still replay its output through a downstream executor. Conversely, an entry may have an exact injective authorization witness even when approvers never communicate. The ANA experiment makes that cross-layer target precise and proves a compiler for it; it does not present ordinary signature unforgeability or peer authentication under a new name.

## Artifact Availability

The four Tamarin case-study theories, rerun scripts, and pinned tool summaries are available at <https://github.com/emiliaprotocol/emilia-protocol>. They are not a mechanized refinement proof of the compiler in this paper. The relevant theories are:

```
formal/tamarin/ep_receipt_core.spthy
formal/tamarin/ep_quorum_core.spthy
formal/tamarin/ep_reliance_composed.spthy
formal/tamarin/ep_six_claim_composed.spthy
```

The v3 predecessor is archived at doi:10.5281/zenodo.21968577. The v4 source and PDF are released with this preprint. Version v4 changes the security target, state experiment, issuance interface, multi-user reduction, nonce model, and prior-work comparison. It is not a cosmetic revision of that record.

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